

2023



Cathay Accelerates Projects and Boosts Reliability by Modernizing Critical Applications on AWS

Cathay modernized 90+ applications on AWS, using Amazon RDS and AWS DMS for seamless database management and migration, and Amazon FSx for NetApp ONTAP to manage IBM MQ workloads, decreasing time-to-market and increasing availability.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)

99.95%

availability for smoother web experien



Hi, I can connect you with an AWS representative or answer questions you have on AWS.



|

Reinforced security

Robust encryption and zero-trust policies



Need more info? Highlight any text to get an explanation generated with AWS generative AI.



reduction in carbon footprint



2

Overview

[Cathay](#) is a premium travel lifestyle brand based in Hong Kong. The business migrated its mission-critical applications to AWS to accelerate time-to-market and improve scalability.

By using [Amazon DMS](#), Cathay migrated over 90 applications in just two years. The company adopted [Amazon Relational Database Service](#) for streamlined database management and [Amazon FSx for NetApp ONTAP](#) to manage IBM MQ workloads. The application modernization project resulted in 37 percent cost savings, enhanced security, and high availability.



Opportunity | Establishing a Foundation for Modern Applications

[Cathay](#) is a premium travel lifestyle brand based in Hong Kong with around 21,000 employees. The Cathay Group provides passenger and cargo services to around 80 destinations worldwide.

Cathay wanted to enhance customer engagement by fostering a DevOps-driven IT culture and adopting cloud computing on Amazon Web Services (AWS). “We recognized the pivotal role of digital transformation in the travel industry and chose AWS to support our transformation,” says Tony Leung, head of IT Platforms at Cathay. In 2021, Cathay began to modernize its applications on the cloud to leverage autoscaling and auto-recovery capabilities. The app modernization project had clear objectives: enhance scalability, increase resilience, and accelerate time-to-market. To achieve these goals, the company's IT team carefully assessed each application, choosing to refactor the majority into containerized microservices.

Solution | Migrating 90+ Applications Seamlessly with Zero Downtime

Cathay collaborated closely with its Solution Architect and Technical Account Manager from [AWS Enterprise Support](#) and the [AWS Infrastructure Event Management](#) team to migrate over 90 applications from on-premises Oracle databases and mainframes to AWS. The migration encompassed critical IT systems, including those utilized for aircraft maintenance and ticketing.

Leung says, “AWS provided ongoing support to enhance our automation and infrastructure as code, including swift issue escalation with dedicated teams offering technical guidance and effective workarounds.” The migration was executed in phases, with a full year of continuous monthly cutover dates carefully coordinated with the company's business units.

Cathay adopted [Amazon Database Migration Service](#) (AWS DMS) to securely migrate its applications with zero downtime. It also relies on [Amazon Relational Database Service](#) (Amazon RDS) to operate and autoscale various database engines. The company is currently transitioning from a traditional database engine to open-source cloud databases, such as PostgreSQL and [Amazon Aurora](#).

Cathay is also one of the first AWS users of [Amazon FSx for NetApp ONTAP](#) as a fully managed shared storage service for its IBM MQ workloads. “Amazon FSx for NetApp ONTAP is easy to use and provides a reliable solution for our high-frequency, high-traffic applications,” Leung shares.

Furthermore, Cathay's security posture has improved by adopting a zero-trust concept, limiting access with dedicated [Amazon Virtual Private Clouds](#) (Amazon VPC) for each application. Additionally, the system encrypts files during transfers between the cloud and on-premises servers, ensuring data remains secure.

The business successfully migrated all targeted applications in two years, staying on budget and on-schedule. Leung says, “The application cutovers were smooth, thanks to all the automations we built into each process. We had tight timelines and hit our goals despite restrictions on resources due to the pandemic, which is a credit to our teams and to AWS.”

Outcome | Delivering New Projects Faster with Automation and Autoscaling



The modernization project streamlined time-to-market through quicker provisioning and autoscaling. By automating activities like system health checks, patching, and server provisioning through infrastructure as code, Cathay experienced a 55 percent reduction in time-to-market, and business units no longer face delays of days or weeks for resource allocation to initiate projects. Leung adds, "Based on feedback from one of our business units, a project that typically requires 18 months for completion was successfully finished in just 9 months. This significant reduction in project duration highlights an improvement in time-to-market."

Through Cathay's collaboration with AWS, it has significantly advanced the team's proficiency in cloud technology and DevOps practices. Leung says, "In the initial stage of our cloud journey, we arranged weekly power sessions for AWS and our internal SMEs to share cloud skills and DevOps practices. These sessions attracted an average of 50 attendees each week, bringing together various IT teams, including infrastructure, security, operations, and development."

Leaders at Cathay are also pleased with how the migration aligns with the company's [sustainable development objectives](#). Leung says, "Sustainability was another key focus area for us and shifting workloads from on-premises infrastructures to AWS has helped us reduce our carbon footprint." In a study conducted by 51 Research, the international analyst firm found that moving on-premises workloads to AWS can lower the workload carbon footprint by nearly 80 percent in APAC.

Since the migration, the business achieved 99.95 percent availability, resolving past server access issues. The stability and flexibility offered by AWS improved customer support, resulting in faster responses and smoother web experiences. "Our teams are excited about the innovative possibilities unlocked by their new DevOps and infrastructure as code skill sets on AWS, enhancing their employability," says Leung.

With nearly 80 percent of its applications on AWS, Cathay is actively planning the adoption of a Multi-Region architecture, further enhancing resiliency and redundancy. It currently utilizes the AWS Asia Pacific (Singapore) Region and plans to deploy in the Hong Kong and Tokyo Regions in the near future. Leung concludes, "The scalability, user-friendliness, and on-demand capabilities of AWS have empowered our team to manage our cloud infrastructure effectively."

About Cathay

Cathay is a premium travel lifestyle brand headquartered in Hong Kong and founded in 1946. Flights are provided by Cathay Pacific, the home airline of Hong Kong, offering both passenger and cargo services worldwide, and is renowned for its quality service, extensive routes, and modern aircraft fleet.

AWS Services Used

Amazon FSx for NetApp ONTAP

Amazon FSx for NetApp ONTAP provides fully managed shared storage in the AWS Cloud with data access and management capabilities of ONTAP.



[Learn more »](#)

AWS Database Migration Service

AWS Database Migration Service (AWS DMS) is a managed migration and replication service that helps move your database and analytics workloads to AWS quickly, securely, and with minimal downtime and zero data loss.

[Learn more »](#)

Amazon Relational Database Service

Amazon Relational Database Service (Amazon RDS) is a collection of managed services that makes it simple to set up, operate, and scale databases in the cloud.

[Learn more »](#)

AWS Countdown

AWS Countdown helps you throughout the project lifecycle to assess operational readiness, identify and mitigate risks, and plan capacity, using proven playbooks developed by AWS experts..

[Learn more »](#)

More Customer Stories

GERMANY



HeyJobs on AWS

HeyJobs aims to make job searching and recruitment more effective for

ITALY



Namirial Streamlines Contract Signing with...



candidates and employers.

2024

Namirial enhanced its eSignAnywhere solution with generative AI using Amazon Bedrock, improving engineering productivity and the user

2024

COSTA RICA



Modernizing Applications in Minutes Using Amazon Q...

Learn how Novacomp accelerated

2024

AUSTRALIA



Dovetail Enhances Customer Insights with...

Learn how Dovetail accelerates development of new generative AI products and features using Amazon Bedrock, deploying the latest large language models while reducing security and administrative overhead.

2024

Get Started

Organizations of all sizes across all industries are transforming their businesses and delivering on their missions every day using AWS. Contact our experts and start your own AWS journey today.

Contact Sales

- What Is AWS?

What Is Cloud Computing?

AWS Accessibility

AWS Inclusion, Diversity & Equity

What Is DevOps?

What Is a Container?

What Is a Data Lake?

What is Artificial Intelligence (AI)?

What is Generative AI?

What is Machine Learning (ML)?

AWS Cloud Security

What's New

Blogs

Press Releases
- Architecture Center

Product and Technical FAQs

Analyst Reports

AWS Partners
- Python on AWS

Java on AWS

PHP on AWS

JavaScript on AWS
- AWS re:Post

Knowledge Center

AWS Support Overview

Legal

AWS Careers



Amazon is an Equal Opportunity Employer: *Minority / Women / Disability / Veteran / Gender Identity / Sexual Orientation / Age.*

- Language
- عربي |

Bahasa Indonesia |

Deutsch |

English |

Español |

Français |

Italiano |

Português |

Tiếng Việt |

Türkçe |

Русский |

ไทย |

日本語 |

한국어 |

中文 (简体) |

中文 (繁體)

